Income Classification and Customer Segmentation for Retail Marketing Strategy

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ABSTRACT

This project addressed two business goals for a retail client using census-based demographic and employment data. First, I built a classification model to predict whether an individual is likely to earn more than $50,000, helping prioritize audiences with greater purchasing power. Second, I developed a customer segmentation framework to group individuals into distinct market segments for more targeted marketing strategies. Because the dataset is highly imbalanced, model evaluation focused on ranking-oriented metrics rather than accuracy alone.

For the classification task, I compared Logistic Regression, Random Forest, and XGBoost using a consistent preprocessing and evaluation pipeline. XGBoost was selected as the final model because it achieved the strongest overall performance, with ROC-AUC of approximately 0.956 and PR-AUC of approximately 0.695, while also providing the best precision-recall balance after threshold tuning. From a business perspective, this model is best used as a prioritization tool to rank prospects by likely income level and allocate premium marketing resources more efficiently.

For the segmentation task, I evaluated multiple clustering approaches and selected a Gaussian Mixture Model with five clusters as the final coarse segmentation because it provided the best balance between clustering quality and business interpretability. The resulting segments reflect meaningful differences in life stage, work status, household structure, and income potential. For example, one segment is dominated by dependents with almost no direct income potential, while another smaller but highly valuable segment shows the strongest concentration of higher-income individuals. This suggests that income prediction alone is not enough for effective targeting; the client should also tailor messaging and offers by segment.

1 BUSINESS OBJECTIVE

The objective of this project is to support a retail client in improving audience targeting and marketing effectiveness using demographic and employment-related data. Specifically, the project focuses on two complementary tasks: predicting whether an individual is likely to belong to the higher-income group and identifying broader customer segments with distinct characteristics.

The first objective, income classification, is intended to help the client prioritize individuals who are more likely to have greater purchasing power. Rather than treating all potential customers equally, the client can use the model to rank and focus on audiences with higher expected value, improving the efficiency of marketing spend and campaign targeting.

The second objective, customer segmentation, is intended to provide a more nuanced understanding of the population beyond a single income label. Even when individuals have similar income levels, they may differ substantially in life stage, household structure, employment patterns, and likely marketing needs. By grouping individuals into interpretable segments, the client can better tailor messaging, offers, and campaign strategies to different types of customers.

Taken together, these two tasks serve different but complementary business purposes: the classification model helps identify who may be more valuable to target, while the segmentation model helps determine how different groups should be approached. This combined framework provides a stronger basis for retail marketing decisions than either approach alone.

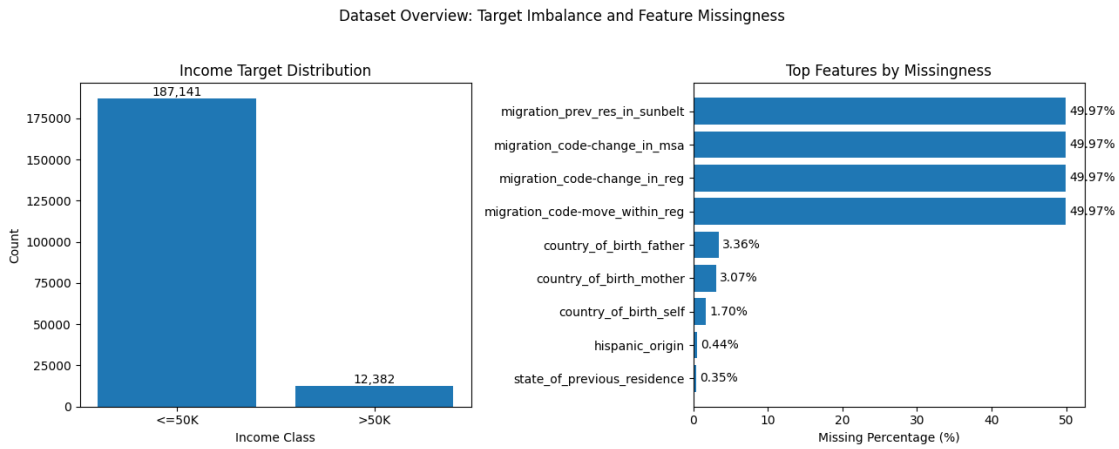


Figure 1. Dataset overview showing the target class imbalance and the top features by missing percentage. These patterns informed the preprocessing design and motivated the use of ranking-oriented evaluation metrics rather than accuracy alone.

2 DATA UNDERSTANDING AND PREPARATION

To better understand the dataset before modeling, I first examined two key characteristics: the distribution of the target variable and the extent of missingness across features. These two aspects are especially important in this project because they directly influence both preprocessing decisions and model evaluation strategy.

The first important observation is that the target variable is highly imbalanced, with most observations belonging to the income group below $50K and only a relatively small portion belonging to the group above $50K. This imbalance implies that classification accuracy alone would not be an informative measure of model quality, since a model could achieve high accuracy by simply predicting the majority class. As a result, later model evaluation emphasized ranking-oriented metrics such as ROC-AUC and PR-AUC, along with threshold tuning to better balance precision and recall.

The second important observation is that several variables contain substantial missingness. In particular, some migration-related fields have a large proportion of missing values, while many categorical variables also include structured categories such as `Not in universe`, which should not be treated the same as random data-entry missingness. These patterns motivated a preprocessing strategy that preserves informative categorical structure while applying robust imputation to handle incomplete records.

To summarize these patterns, Figure 1 presents a combined overview of the dataset, including the target distribution and the top features by missing percentage. Together, these patterns show that the dataset requires careful treatment of imbalance and missing data before any meaningful classification or segmentation analysis can be performed.

3 INCOME CLASSIFICATION

The objective of the classification task is to identify individuals who are more likely to belong to the above-$50K income group. From a business perspective, this model is intended to support audience prioritization rather than serve as a rigid approval-or-rejection rule. By estimating the likelihood that a prospect belongs to a higher-income segment, the client can allocate premium marketing resources more efficiently and focus outreach on individuals with greater expected purchasing power.

3.1 Modeling Approach

To develop a robust classification solution, I compared three candidate models: Logistic Regression, Random Forest, and XGBoost. This comparison was designed to balance interpretability, modeling flexibility, and predictive strength.

Logistic Regression was included as a transparent baseline. Its main advantage is interpretability, since it provides a relatively simple and well-understood relationship between features and the predicted outcome. However, because the dataset contains many categorical variables and potentially complex nonlinear interactions, Logistic Regression may be too restrictive to fully capture the structure of the problem.

Random Forest was included as a nonlinear ensemble benchmark. It can capture feature interactions and nonlinear patterns

Table 1. Classification model comparison across candidate models.

Model	ROC-AUC	PR-AUC	Best Threshold	Best F1	Precision	Recall
Logistic Regression	0.9458	0.6122	0.2692	0.5794	0.5485	0.6139
Random Forest	0.9498	0.6541	0.2475	0.6025	0.5713	0.6373
XGBoost	0.9561	0.6951	0.2930	0.6305	0.6163	0.6454

Table 2. Sensitivity analysis for XGBoost under different `min_frequency` settings.

Setting	ROC-AUC	PR-AUC	Best Threshold	Best F1	Precision	Recall
XGBoost (`min_frequency = 0.01`)	0.9561	0.6951	0.2930	0.6305	0.6163	0.6454
XGBoost (`min_frequency = 0.005`)	0.9560	0.6948	0.3505	0.6335	0.6572	0.6116

more effectively than Logistic Regression and is generally robust to noise. However, in a high-dimensional sparse setting created by one-hot encoding, Random Forest may not always deliver the strongest ranking performance.

XGBoost was included as the most flexible tabular learning candidate. Gradient boosting methods are often highly effective for structured data because they can capture nonlinear relationships, interactions, and heterogeneous decision boundaries more efficiently than simpler models. Although XGBoost is less interpretable than Logistic Regression and requires more tuning, it is often one of the strongest choices for tabular classification tasks.

3.2 Evaluation Strategy

Because the target variable is highly imbalanced, classification accuracy alone would not provide a meaningful view of model quality. A naive model could achieve high accuracy by predicting the majority class most of the time while still failing to identify the smaller but more business-relevant higher-income group.

For this reason, model evaluation focused on ROC-AUC and PR-AUC. ROC-AUC measures the model's overall ability to rank positive observations above negative ones across thresholds, making it a useful measure of general discrimination. PR-AUC is particularly important in this setting because it focuses on precision and recall for the positive class, which is more informative when the positive class is relatively rare.

In addition to these threshold-independent metrics, I evaluated the best F1 score obtained by tuning the decision threshold. This was important from a business perspective because the client may need different precision-recall tradeoffs depending on campaign cost, available budget, and tolerance for false positives.

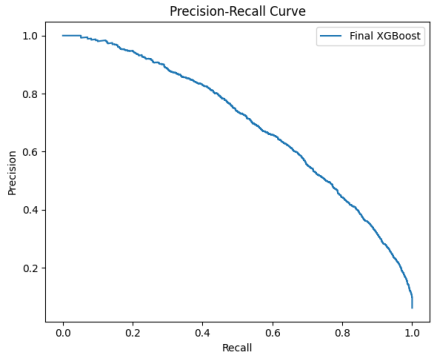
3.3 Results and Final Model Selection

Table 1 summarizes the performance of the three candidate models. Among them, XGBoost achieved the strongest overall results, with the highest ROC-AUC, the highest PR-AUC, and the best F1 score after threshold tuning. This indicates that XGBoost provided the best balance between general ranking performance and positive-class retrieval quality.

To further assess the stability of the selected model, I tested XGBoost under different category-grouping settings through the `min_frequency` parameter. As shown in Table 2, the model remained highly stable across the two settings. When `min_frequency = 0.005`, ROC-AUC remained nearly unchanged while the best F1 score improved slightly from 0.6305 to 0.6335. Based on this result, I selected the `min_frequency = 0.005` XGBoost configuration as the final classifier.

Figure 2 presents the Precision–Recall curve for the final XGBoost model. This is the most informative visual for the main text because it directly shows how precision declines as recall increases, which is especially relevant for an imbalanced classification problem. The curve confirms that the model maintains strong precision over a broad range of recall values, supporting its suitability for ranking and targeting tasks.

Figure 3 provides two additional views of final-model performance: the ROC curve and the normalized confusion matrix. The ROC curve confirms strong overall discrimination, with an AUC of approximately 0.956. The confusion matrix shows that at the selected operating threshold, the model correctly classifies approximately 97.3% of the below-$50K group and 65.6% of the above-$50K group. Together, these results indicate that the final classifier is highly effective at separating the two classes while still recovering a substantial share of the minority higher-income group.



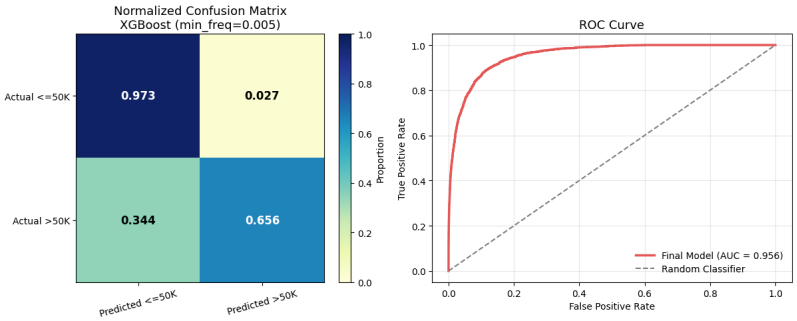


Figure 2. Precision–Recall curve for the final XGBoost classifier.

Figure 3. ROC curve and normalized confusion matrix for the final XGBoost classifier.

Overall, XGBoost was selected as the final classifier because it consistently outperformed the other models across the most relevant evaluation criteria. It not only demonstrated the strongest discrimination ability, but also provided the best precision–recall tradeoff for identifying the higher-income group.

3.4 Business Interpretation and Deployment Considerations

From a business perspective, the classification model is most useful as a scoring and prioritization tool. A higher predicted probability indicates that an individual is more likely to belong to the above-$50K group and may therefore be a stronger candidate for premium offers, higher-value campaigns, or more personalized outreach. Rather than using the model as a strict binary decision rule, the client should use it to rank prospects and allocate resources according to expected value.

This model is especially useful in settings where the client has limited behavioral or transactional data and needs to rely on demographic and employment-related variables as proxies for economic potential. In those scenarios, the model can provide a practical first-pass targeting layer. However, it is important to recognize its limitations. Income is only a proxy for purchasing power, and demographic data cannot fully capture actual shopping behavior, product preferences, or campaign responsiveness. In addition, the threshold that defines a positive prediction should not be fixed universally. The optimal threshold depends on campaign economics: if outreach is inexpensive, the client may prefer a lower threshold to maximize reach; if premium targeting is costly, a higher threshold may be more appropriate to improve precision.

Given these constraints, the recommended deployment strategy is to use the model in a staged targeting workflow. First, score all prospects using the classification model. Second, rank individuals based on predicted probability rather than immediately converting predictions into hard labels. Third, apply different score cutoffs depending on campaign objectives and budget constraints. Finally, combine the classification score with the segmentation results so that the client can not only identify who is more likely to be high income, but also tailor the message and offer according to segment characteristics.

4 SEGMENTATION MODELING

The purpose of the segmentation task is to identify meaningful subgroups within the population that can support differentiated marketing strategies. Unlike the classification model, which estimates whether an individual is likely to belong to the above-$50K income group, the segmentation model is designed to explain how individuals differ in life stage, labor-force attachment, household structure, and financial profile. This makes the segmentation output more suitable for campaign design, message tailoring, and audience prioritization.

4.1 Feature Design and Preprocessing

For clustering, I excluded the income label, binary target, and survey weight from the feature space so that the segmentation would reflect underlying customer structure rather than the prediction target itself. The clustering pipeline instead used demographic, employment, household, and financial variables, including continuous features such as age, wage per hour, capital gains, capital losses, dividends from stocks, and weeks worked in year, together with a large set of categorical census variables.

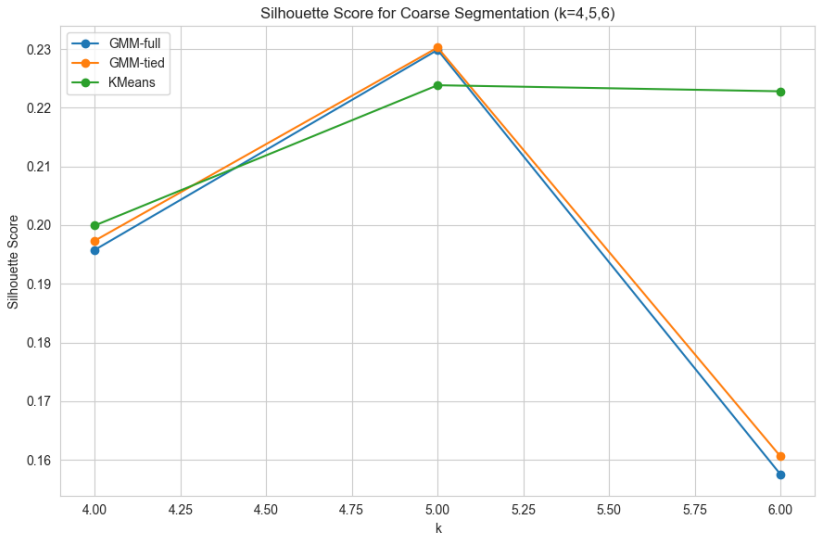


Figure 4. Silhouette score comparison across candidate coarse segmentation models for $k = 4, 5, 6$. Performance peaks around $k = 5$, supporting the choice of a five-cluster solution.

Because the encoded feature matrix becomes high-dimensional and sparse after one-hot encoding, I applied Truncated SVD before clustering. This step compresses the mixed numeric–categorical feature space into a lower-dimensional latent representation while preserving most of the main structure. It also improves clustering stability and makes the segmentation easier to visualize and explain.

4.2 Candidate Models and Selection Logic

I evaluated multiple coarse segmentation approaches, including KMeans and Gaussian Mixture Models, across different values of k. Since segmentation is an unsupervised task, model selection cannot rely on predictive accuracy. Instead, I used internal clustering metrics—especially the silhouette score—together with business interpretability.

Figure 4 summarizes the model-selection evidence. The strongest configurations occurred around $k = 5$, while the $k = 6$ solutions showed a noticeable decline in silhouette performance for the Gaussian Mixture variants. Although KMeans also performed competitively, I selected a **Gaussian Mixture Model with 5 clusters** as the final coarse segmentation because it provided a strong balance between statistical separation, softer cluster assignment, and business interpretability.

4.3 Final Coarse Segmentation

Figure 5 shows the final five-cluster Gaussian Mixture segmentation in a two-dimensional embedding. This figure should be interpreted as a simplified communication view rather than a complete representation of cluster quality. The clustering model is fit in the higher-dimensional SVD feature space, so some visible overlap is expected when the latent structure is compressed into two dimensions.

For that reason, the apparent overlap in the 2D plot should not be interpreted as evidence that the final model is weak. Instead, the five-cluster solution should be viewed as a deliberate coarse segmentation that captures the major structural differences in the population while remaining compact enough for business use.

4.4 Exploratory Fine-Grained Structure

To complement the final coarse segmentation, I also generated an exploratory UMAP+HDBSCAN visualization. Figure 6 suggests that the population contains additional fine-grained local structure beyond the chosen five-cluster solution. In other words, the data could be partitioned more aggressively into smaller subgroups.

However, finer segmentation is not automatically better for business use. A highly fragmented solution may reveal local structure, but it is harder to explain, harder to operationalize, and less practical for downstream campaign design. For that reason,

Figure 5. Final coarse segmentation produced by the Gaussian Mixture Model with $k = 5$. Although some overlap remains in the 2D projection, the model is fit in a higher-dimensional latent space and is selected because it provides a more compact and business-usable segmentation.

Table 3. Summary of final coarse segments from the Gaussian Mixture model.

Cluster	Share	High-income rate	Age mean	Weeks worked mean	Wage/hr mean	Capital gains mean	Dividends mean
0	27.35%	0.00%	8.17	0.93	0.00	0.00	1.44
1	3.70%	32.70%	48.50	39.45	77.57	11759.23	1028.48
2	31.79%	7.09%	44.76	29.83	0.00	0.00	245.89
3	5.44%	4.43%	36.52	45.00	967.05	0.03	88.32
4	31.72%	7.89%	44.92	30.05	0.00	0.00	240.02

I retained the Gaussian Mixture model with $k = 5$ as the final *operational* segmentation and used the UMAP+HDBSCAN result only as an exploratory reference.

4.5 Cluster Profile Summary

To translate the segmentation into a business-usable framework, Table 3 summarizes the size and numeric profile of each final cluster. The five clusters differ meaningfully in age, labor-force activity, wage profile, asset-related signals, and concentration of higher-income individuals.

4.6 Interpretation of the Final Segments

The final five-cluster solution can be interpreted as a set of broad customer archetypes.

Cluster 0: Dependents / Youth. This segment is very young on average, shows almost no labor-force participation, and has essentially no direct high-income potential. From a business perspective, this group has little standalone premium targeting value and is better viewed as a dependent population rather than an independent customer segment.

Cluster 1: Affluent Asset-Rich Adults. This is the smallest but most valuable segment. It has by far the highest high-income rate, the strongest capital-gain and dividend profile, and a relatively mature age structure. This segment is the clearest premium audience and is the most relevant group for high-value products, wealth-oriented campaigns, or resource-intensive outreach.

Figure 6. Exploratory UMAP+HDBSCAN visualization showing finer-grained local structure in the population. This figure is included as a supporting reference rather than the final operational segmentation.

Cluster 2: Mainstream Mature Households. This is one of the largest clusters and represents a broad adult population with moderate labor-force attachment and nontrivial high-income potential. It appears to capture a mainstream household segment suitable for general retail targeting and broad campaign coverage.

Cluster 3: High-Intensity Wage Earners. This segment is smaller but distinct. It has the strongest wage-per-hour signal and high weeks-worked values, but much weaker asset-related signals than Cluster 1. This suggests a profile of actively employed earners rather than asset-rich households. It may respond better to work-related or income-driven positioning than to wealth-oriented messaging.

Cluster 4: Broad Middle-Market Households. This cluster is similar in size to Cluster 2 and has a broadly adult, household-oriented profile. Its high-income rate is somewhat stronger than Cluster 2, but far below Cluster 1. It appears to be a useful broad-reach segment for mainstream offers and family-oriented campaigns.

4.7 Important Features Influencing Segmentation

To make the segmentation more interpretable, I examined how key categorical variables align with the low-dimensional structure. Figure 7 shows the same 2D projection colored by employment status, class of worker, and tax filer status. These overlays indicate that the final segmentation is strongly associated with labor-force attachment, work arrangement, and filing or household structure.

More broadly, the clustering appears to be driven by five interpretable dimensions:

- **Life stage**, especially age;
- **Labor-force attachment**, especially weeks worked and employment status;
- **Employment arrangement**, especially class of worker;
- **Household and filing structure**, especially tax filer status;
- **Asset intensity**, especially capital gains and dividends from stocks.

These dimensions are important because they are not merely statistical separators; they also translate naturally into marketing language. In other words, the clusters differ not only numerically, but also in ways that support differentiated business action.

Figure 7. Two-dimensional projection colored by selected categorical variables. These overlays suggest that employment status, class of worker, and tax filing structure are important drivers of the observed segmentation pattern.

4.8 Business Interpretation

From a business perspective, segmentation adds a layer that the classification model alone cannot provide. Two individuals may have similar predicted probabilities of being above $50K, but belong to very different segments—for example, an affluent asset-rich adult versus an actively employed wage earner. These cases may require different products, different value propositions, and different messaging styles.

The recommended workflow is therefore to combine the two outputs: first use the classification model to prioritize likely higher-income prospects, and then use segment identity to decide how to engage them. This creates a more realistic targeting framework in which the client can prioritize both *who* to target and *how* to market to them.

5 LIMITATIONS AND NEXT STEPS

5.1 Limitations

First, the data is demographic and employment-oriented rather than behavioral. That makes it useful for broad targeting, but it does not capture the kinds of signals that are often most predictive in a retail setting, such as prior purchase behavior, recency, frequency, channel preference, or product affinity. For that reason, these models should be viewed as a strong starting point rather than a complete customer decisioning system.

Second, the classification model is best interpreted as a ranking tool, not as a literal measure of future customer value. Income is only one dimension of commercial potential, and the above-$50K label is an imperfect proxy for retail opportunity.

Third, the segmentation results are intentionally coarse. The five-cluster solution was chosen because it is easier to explain and more practical to use, not because it captures every nuance in the data. The exploratory UMAP view suggests that more granular structure exists, but a more detailed segmentation would also be harder to operationalize consistently.

Finally, some evaluation choices remain context-dependent. The selected threshold is a reasonable technical default, but the right operating point will depend on actual campaign costs, channel economics, and the client's tolerance for false positives.

5.2 Next steps

The most valuable next step would be to test these models against real business outcomes. If campaign response, conversion, or spend data becomes available, the client should validate whether higher model scores and segment-specific treatments actually produce better commercial results.

A second priority would be to enrich the feature set with customer behavior data. Adding transaction history, channel engagement, browsing behavior, or product-level interactions would likely improve both targeting precision and segment relevance.

A third next step would be to formalize deployment rules. Rather than relying on a single universal threshold, the client could define different score cutoffs by campaign type, budget level, or objective. This would make the system more flexible and more aligned with business constraints.

Finally, the segmentation could be revisited after deployment. Once the client observes how the initial five segments perform in practice, it may make sense to either simplify them further for execution or expand certain high-value groups into more detailed subsegments.

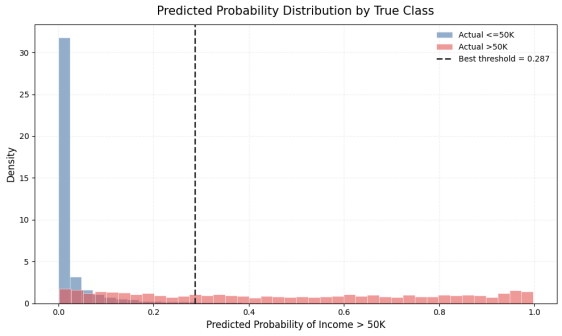


Figure A-1. Predicted probability distribution by true class, with the selected threshold shown by the dashed line.

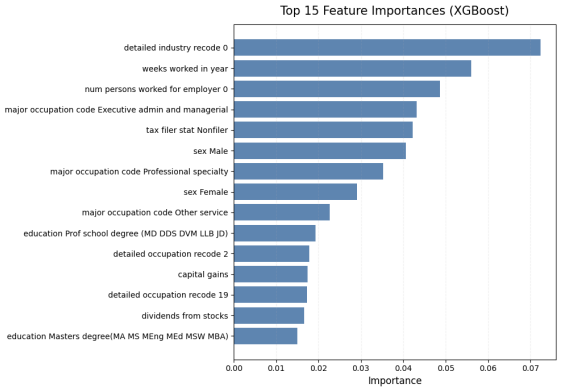


Figure A-2. Top 15 feature importances from the final XGBoost classifier.

6 APPENDIX

This appendix provides a compact set of supporting materials for the final report. It includes two diagnostic figures for the classification model and two summary tables that support model comparison and segment interpretation.

6.1 Additional Classification Diagnostics

Figure A-1 shows the distribution of predicted probabilities by true class, with the selected operating threshold marked by the dashed line. This figure helps explain why the final classifier is better used as a prioritization tool than as a rigid yes-or-no rule.

Figure A-2 reports the top feature importances from the final XGBoost model. These variables help clarify which parts of the profile are most influential in distinguishing the higher-income group.

6.2 Supporting Tables

Table 4 summarizes the main missing-value and placeholder-value patterns observed during the data audit. The most important takeaway is that many fields do not contain ordinary missingness in the usual sense. Instead, they frequently use structured placeholder categories such as `not in universe`, which carry substantive meaning in census-style data. For that reason, these values were generally preserved as valid categories rather than treated as data-entry errors or dropped records. This distinction is especially important for employment, migration, and household-related variables, where placeholder values often reflect life-stage or eligibility conditions rather than incomplete collection.

Table 4. Summary of missing-value and placeholder-value patterns in selected fields

Column	Unique values	Most common values	Placeholder / suspicious values noted
state_of_previous_residence	50	Dominated by `not in universe`	`not in universe`
country_of_birth_mother	42	Mostly `united-states`, followed by `mexico`	None flagged
country_of_birth_father	42	Mostly `united-states`, followed by `mexico`	None flagged
country_of_birth_self	42	Mostly `united-states`, followed by `mexico`	None flagged
major_industry_code	24	`not in universe or children` is dominant	`not in universe or children`
major_occupation_code	15	`not in universe` is dominant	`not in universe`
class_of_worker	9	`not in universe`, then `private`	`not in universe`
full_or_part_time_employment_stat	8	`children or armed forces` is dominant	None flagged
migration_code-change_in_msa	9	`nan`, `nonmover`	`not in universe`
migration_code-move_within_reg	9	`nan`, `nonmover`	`not in universe`
migration_code-change_in_reg	8	`nan`, `nonmover`	`not in universe`
migration_prev_res_in_sunbelt	3	`nan`, `not in universe`, `no`	`not in universe`
region_of_previous_residence	6	Dominated by `not in universe`	`not in universe`
reason_for_unemployment	6	Dominated by `not in universe`	`not in universe`
family_members_under_18	5	Dominated by `not in universe`	`not in universe`
live_in_this_house_1_year_ago	3	Includes `not in universe under 1 year old`	`not in universe under 1 year old`
enroll_in_edu_inst_last_wk	3	Dominated by `not in universe`	`not in universe`
member_of_a_labor_union	3	Dominated by `not in universe`	`not in universe`
fill_inc_questionnaire_for_veterans_admin	3	Dominated by `not in universe`	`not in universe`

Table 5. Final cluster summary and business interpretation

Cluster	Share	High-income rate	Quantitative profile	Dominant characteristics	Business interpretation
0	27.35%	0.00%	Mean age = 8.17, mean weeks worked = 0.93, wage income and capital gains are essentially zero	Overwhelmingly children, never married, nonfilers, and "not in universe" / child-related labor-force categories	**Dependents / youth**. This segment has almost no direct income potential and should generally be excluded from premium targeting. If retained, it should only be used as context within a broader household-level strategy.
1	3.70%	32.70%	Mean age = 48.50, mean weeks worked = 39.45, highest capital gains and dividends, strongest asset profile	More educated, more likely married, more likely joint filers, and more represented in private or self-employed work	**Affluent asset-oriented households**. This is the highest-value segment in the population. It is the strongest candidate for premium offers, financial products, and campaigns aimed at higher purchasing power.
2	31.79%	7.09%	Mean age = 44.76, mean weeks worked = 29.83, little wage or capital-gains activity, moderate dividends and capital losses	Broad mainstream education mix, married-heavy, mixed labor attachment, substantial private-sector representation	**Mainstream mixed households**. This is a large, middle-profile audience with modest but meaningful income potential. It is suitable for broad household campaigns and mid-tier targeting rather than highly selective premium outreach.
3	5.44%	4.43%	Mean age = 36.52, mean weeks worked = 45.00, very high wage-per-hour values, but little asset income	Primarily private-sector workers, strongly attached to work, but less asset-backed and less concentrated in high-income status	**Wage-driven working adults**. This segment appears highly work-attached but not especially wealthy on broader financial measures. It may respond better to practical, income-linked, or value-oriented messaging than to asset-oriented premium positioning.
4	31.72%	7.89%	Mean age = 44.92, mean weeks worked = 30.05, minimal wage and capital-gains values, moderate dividends and capital losses	Similar to Cluster 2 in age and household structure, but with stronger concentration in full-time schedules and labor-force participation	**Working mainstream households**. This segment looks like an actively employed mainstream audience. Relative to Cluster 2, it is somewhat more attached to regular work and may be a better target for general retail or household campaigns with moderate value emphasis.